

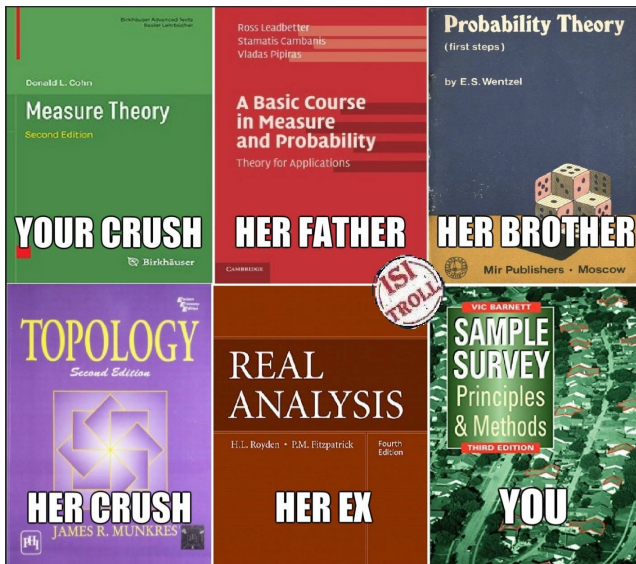
Sample Surveys

Class 1

Ambarish Chattopadhyay

Indian Statistical Institute, Kolkata

Fall 2026



§ 2.1 DEFINITION OF STATISTICS

The word *statistics* is associated with the term *state* (indicating quantitative information relating to state administration). The term *statistics* was coined by the German scholar Gottfried Achenwall around the middle of the eighteenth century. In modern age, this word includes all branches of human activity concerned with numerical data.

Statistics is used as a plural noun and also as a singular noun.

In plural sense, numerical data, i.e., numerical statements of facts in connection with any field of human experience are called *statistics*. For example, production statistics of a factory over different years, statistics of birth and death of a country, etc.

The term *statistics*, in singular sense, means the aggregate of scientific methods which deals with the **collection**, presentation, analysis and interpretation of numerical data. This singular sense of *statistics* is used in the statement, "R. A. Fisher had contributed a lot in various branches of statistics".

- Design of Experiments & Sample Surveys:
 - Deals with the **collection** of data, and more...

Sample Surveys: ISI's DNA



- Mahalanobis's work in large-scale sample surveys in the 1930s and 40s put ISI Kolkata on the global map.
- Estimating Bengal's massive [jute production](#).
 - Sample survey was cheaper, faster, and more accurate than full government census.
- Birth of [National Sample Survey](#) (NSS, 1950).
 - The most comprehensive, multi-subject, continuous household survey in the world.
 - Template for the United Nation's global sampling guidelines.
- Theory and methods born from [practice](#).
 - Rigorous constrained optimization problems.
 - Interpenetrating Network of Sub-sampling (IPNS): Doing resampling before it was cool.
 - Pilot surveys: Doing sequential learning before it was cool.

Questions



- How much does a typical family in a district spend each month on food, rent, and basic needs?
- What fraction of young adults in a state are currently working or looking for work?
- What is the average birth weight of newborns delivered in public hospitals across a state?
- What portion of a city's population developed antibodies after a virus outbreak?
- How many tigers or elephants are currently living in a national park reserve?
- What fraction of micro-transactions on a UPI app fail or run into errors on a given day?
- What is the average distance a student in a rural area has to walk to reach the nearest high school?
- Who do voters plan to support in an upcoming election?

Marks distribution

- Midsem: 30%.
- Endsem: 50%.
- Final presentations: 20%.
 - Students divided into groups.
 - Topics not directly covered in class.
- Google classroom?

The ABCD of Sample Surveys

Example

The state health ministry is planning to launch a new health insurance scheme. To budget for this, they must estimate the **average monthly out-of-pocket medical expenditure** for **all adults in West Bengal**.

- **Observation unit:** An object on which a measurement is taken, e.g., an adult person.
 - Could be anything: an animal, a region, a hospital, a plot of agricultural land, a financial record.
- **Target population:** The ideal, complete collection of observation units (in a particular time period) we want to infer about, e.g., all adults in West Bengal.

Survey by *complete enumeration* or *census*: study everybody

- Collect data on one or more variables of interest (e.g., average monthly out-of-pocket medical expenditure) for **all the units** in the population.
- Quite **hard**, **expensive**, and **time consuming** if the population is big.

Alternative option: Survey by *sampling*

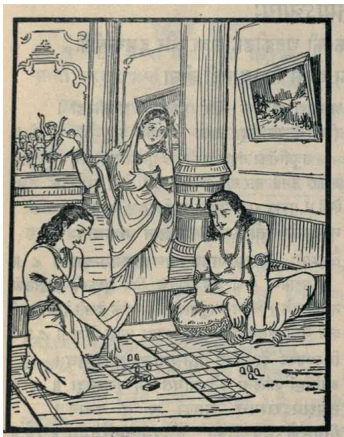
The ABCD of Sample Surveys

- **Sample:** A subset of a population.
- **Sampled population:** The collection of all possible observation units that might have been chosen in a sample, i.e., the part of the target population that is accessible.
 - In an ideal survey, sampled population = target population.
 - In our example, it could be those adults living in registered households.
- **Sampling unit:** A unit that can be selected for a sample.
 - If we don't have the list of adults, households could serve as the sampling units, and the observation units are the adults living in the households.
- **Sampling frame:** The tangible list, map, database, or other specification of sampling units from which a sample may be selected.
 - E.g., the most recent West Bengal Electoral Roll.
 - Must have unique IDs or serial numbers.
 - Proper care should be taken to create the sampling frame.

Sample survey

The process of selecting a sample, and collecting data on one or more variables of interest for these selected units, to estimate characteristics of the target population.

Earliest instance of sampling



- *Mahabharata*: Story of King Nala and Rtuparna.
- Nala loses his kingdom in gambling, wanders in the forest, takes a job as a charioteer to Ayodhya king Rtuparna.
- On a journey, they pass a sprawling tree. By glancing at the tree, he tells Nala the total number of fruits and leaves on the entire tree.
- Nala chops down the tree and spends the whole night counting every single leaf and fruit. He gets amazed by the accuracy of Rtuparna's guess.
- Rtuparna reveals his secret: he gives his estimate based on examining one small branch.

"Know me to be a knower of the secret of the dice and therefore adept in the art of enumeration."

Bad sample: The *Literary Digest* case

- One of the **worst** political predictions in history.
- Who will win the 1936 US presidential election?
 - Franklin Roosevelt (**democratic**) vs Alfred Landon (**republican**)
 - 10 million prospective voters were sent questionnaires.
 - The poll predicted that Landon would win.

The Literary Digest
NEW YORK OCTOBER 31, 1936

Topics of the day

LANDON, 1,293,669; ROOSEVELT, 972,897
Final Returns in The Digest's Poll of Ten Million Voters

Well, the great battle of the ballots in the Poll of ten million voters, scattered throughout the forty-eight States of the Union, is now finished, and in the table below we record the figures received up to the hour of going to press.

These figures are exactly as received from more than one in every five voters polled in our country—they are neither weighted, adjusted nor interpreted.

Never before in an experience covering more than a quarter of a century in taking polls have we received so many different varieties of criticism—praise from many; condemnation from many others—and yet it has been just of the same type that has come to us every time a Poll has been taken in all these years.

A telegram from a newspaper in California asks: "Is it true that Mr. Hearst has purchased *The Literary Digest*?" A telephone message only the day before these lines were written: "Has the Repub-

lican National Committee purchased *The Literary Digest*?" And all types and varieties, including: "Have the Jews purchased *The Literary Digest*?" "Is the Pope of Rome a stockholder of *The Literary Digest*?" And so it goes—all equally absurd and amusing. We could add more to this list, and yet all of these questions in recent days are but repetitions of what we have been experiencing all down the years from the very first Poll.

Prediction—Now, are the figures in this Poll correct? In answer to this question we will simply refer to a telegram we sent to a young man in Massachusetts the other day in answer to his challenge to us to wager \$100,000 on the accuracy of our Poll. We wired him as follows:

"For nearly a quarter century, we have been taking Polls of the voters in the forty-eight States, and especially in Presidential years, and we have always merely mailed the ballots, counted and recorded those returned and let the people of the Nation draw their conclusions as to our accuracy. So far, we have been right in every Poll. Will we be right in the current Poll? That, as Mrs. Roosevelt said concerning the President's reelection, is in the 'lap of the gods.'"

"We never make any claims before election but we respectfully refer you to the opinion of one of the most quoted citizens to-day, the Hon. James A. Farley, Chairman of the Democratic National Committee. This is what Mr. Farley said October 14, 1936:

"Any sane person can not escape the implication of such a gigantic sampling of popular opinion as is embraced in *The Literary Digest* straw vote. I consider this conclusive evidence as to the desire of the people of this country for a change in the National Government. *The Literary Digest* poll is an achievement of no little magnitude. It is a Poll fairly and correctly conducted."

In studying the table of the voters from —

The statistics and the material in this article are the property of J. P. & W. W. W. Co. and are not to be reproduced in any form without the special permission of the copyright owner.

Bad sample: The *Literary Digest* case



- The prediction was a disaster: Roosevelt beat Landon in a landslide victory.
- *Literary Digest* magazine became a laughing stock.
- Magazine got shut down in 1938, never to return again.

What went wrong?

SECRET BALLOT—No Signature—No Condition—No Obligation—Just Mark Your Choice—Mail at Once

CANDIDATES FOR PRESIDENT OFFICIALLY NOMINATED
(Names Arranged Alphabetically)

Put a Cross ☒ in Square Before the Name of Presidential Candidate You Prefer

☐ John W. Landon (Republican)	☐ Franklin D. Roosevelt (Democratic)
☐ Earl Browder (Socialist)	☐ Norman Thomas (Socialist)
☐ Leigh Colvin (Prohibitionist)	☐
☐ Alfred M. Landon (Republican)	☐

Mark How You Voted For President in 1932

☐ Roosevelt
☐ Landon
☐ Clegg
☐ Hoover
☐ Smith
☐ Clegg
☐ Hoover
☐ Smith

Did Not Vote:
☐ Under Legal Age
☐ Other Reasons

(This is important and will reveal the significant shifts from one party to another.)

To assist in tabulation please write name of your State here: _____

The *Literary Digest* survey

What went wrong?

- Potential voter names were compiled from sources such as **telephone directories** and **vehicle registration lists**.
 - These lists favored rich voters who leaned republican (Landon), leaving out poorer working-class citizens who supported Roosevelt.
- Many of the voters **did not respond**.
 - 2.3 million people actually responded. (still a pretty large sample)
 - Persons supporting Landon were **much more likely to respond** to the survey.
 - In fact, many Roosevelt supporters did not remember receiving a survey.

Bad sample: The *Delphi-Facebook* survey

- What percentage of the US adult population is vaccinated against COVID-19?
- Survey done by Delphi Research Group at Carnegie Mellon University, in partnership with Facebook.

nature

[Explore content](#) ▾ [About the journal](#) ▾ [Publish with us](#) ▾

[nature](#) > [articles](#) > [article](#)

Article | Published: 08 December 2021

Unrepresentative big surveys significantly overestimated US vaccine uptake

[Valerie C. Bradley](#), [Shiro Kuriwaki](#), [Michael Isakov](#), [Dino Sejdinovic](#), [Xiao-Li Meng](#) & [Seth Flaxman](#) 

Nature **600**, 695–700 (2021) | [Cite this article](#)

 [Save article](#)

109k Accesses | 224 Citations | 1503 Altmetric | [Metrics](#)

Abstract

Surveys are a crucial tool for understanding public opinion and behaviour, and their accuracy depends on maintaining statistical representativeness of their target populations by minimizing biases from all sources. Increasing data size shrinks confidence intervals but

*“We show how a survey of 250,000 respondents can produce an estimate of the population mean that is **no more accurate than an estimate from a simple random sample of size 10**. Our central message is that **data quality matters more than data quantity**, and that compensating the former with the latter is a mathematically provable losing proposition.”*

- A massive 2021 study capturing roughly 250,000 weekly responses via Facebook.
- Facebook users opting into health surveys were **systematically more likely** to be vaccinated than the general public.
- Despite an enormous sample size, the survey **overestimated** actual US vaccine uptake by a massive 17 percentage points.
- The “Big data paradox.”

Importance of *survey design*

More data $\nRightarrow$ Better data

- Need to make sure that the sample selected is **representative** of the target population.
- A large **unrepresentative** sample may do **more damage** than a small one.
- Past accuracy of a flawed sampling procedure **does not guarantee** future results.
 - The *Literary Digest* surveys worked well for the previous five successive elections.
 - Need to have statistically sound sampling and estimation procedures.
 - Otherwise, disasters can happen anytime.

Good sample



- A sample is **representative** if it can be used to “**reconstruct**” what the population looks like
 - and if we can provide an **accurate assessment** of how good that reconstruction is.
- Sample survey theory provides the only mathematically rigorous foundation for ensuring population representativeness.
- The **design** of the survey, i.e., **how the units are selected**, is very important.

Good sample

Objection to sampling

- “My opinion has **never been asked**, so how can the survey results claim to represent **me**?”
- “Extrapolating what tens of millions are thinking from a tiny sample of opinions **insults** human intelligence and **negates** true freedom of thought.”

What can sampling buy us?

- **Reduced** cost, manpower, time, and **enlarged** scope.
- **Easier** to manage organizational work. Can use a small number of highly trained people.
- Estimates based on sample surveys are often **more accurate** than those based on a census.

$$\text{Total error} = \underbrace{\text{Sampling error}}_{\text{Error due to examining only a part of the population}} + \underbrace{\text{Nonsampling error}}_{\text{Measurement errors, missing data, poorly trained investigators}}$$

Different types of sampling

Probability sampling or Random sampling

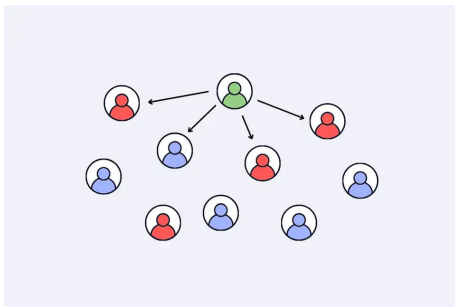
- The selection of units in the sample from a population is governed by a **mathematical randomization mechanism** (based on laws of chance or probability).
- Every unit in the sampling frame has a **known, strictly positive probability** of being selected.
- Can use this randomization mechanism to do **inference**.

Non-probability sampling or Non-random sampling

- The selection of units in the sample from a population is **not governed** by the probability laws.
- **When it is used:** Exploratory research, qualitative studies, pilot testing, or situations where constructing a comprehensive sampling frame is impossible.
- Hard to do statistical inference.

Non-random sampling: Some special cases

Convenience sampling:



- Units are included simply because they are **immediately accessible and cheap to reach** for the researcher.
- **When it is used:** Quick pilot tests to check if survey wording makes sense.
- E.g., a researcher handing a draft questionnaire to people in the university canteen.

Non-random sampling: Some special cases

Judgment (or purposive) sampling:



- The researcher uses their own domain expertise to **hand-pick specific units** they believe will yield the most informative or relevant data.
- **When it is used:** Highly specialized case studies or expert interviews.
- E.g., a political analyst assessing a government's performance by intentionally interviewing only local "buddhijibis" (intellectuals).

Non-random sampling: Some special cases

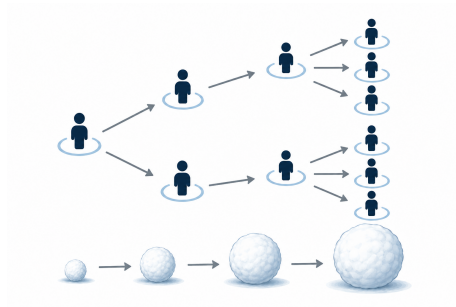
Self-selection sampling:



- Individuals **voluntarily opt-in** to participate in the study entirely on their own initiative.
- **When it is used:** Customer feedback forms, informal internet/radio polls, or psychological studies requiring highly motivated volunteers.
- The statistics from such surveys **cannot be trusted**. At best, they are entertainment; at worst, they mislead.

Non-random sampling: Some special cases

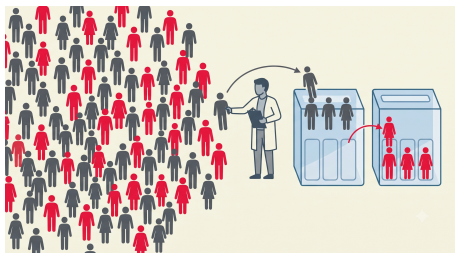
Snowball sampling:



- The researcher selects one person (or a few people) for the study. These individuals are asked to **recruit or refer other individuals** from their personal networks who fit the study criteria.
- **When it is used:** Reaching hidden, rare, or vulnerable populations that lack a formal sampling frame.
- E.g., getting a list of patients having HIV/AIDS in a city may be difficult (because most of such persons won't want to be found). Here, snowball sampling can be used.

Non-random sampling: Some special cases

Quota sampling:



- The researcher sets a **specific target number** for different groups (like exactly 50 men and 50 women), and then fills those spots using convenience sampling until the target is met.
- **When it is used:** When you need a guaranteed mix of specific groups quickly and cheaply, without the effort or cost of actual random selection.
- E.g., a student wants feedback on the campus Wi-Fi. She sets a quota to interview exactly 25 first-year students and 25 second-year students and surveys students in the canteen until she hits the numbers.